

DILR PAPER – 2

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

The table gives the percentage growth in sales of five companies, over the respective previous years.

Company	2016	2017	2018	2019
P	10	5	15	10
Q	10	10	10	20
R	20	10	0	15
S	5	20	5	5
T	10	15	5	25
U	25	0	5	30

Which company had the highest percentage increase in sales from 2015 to 2018?

☐ a) P

☒ b) Q (Your answer - ✓)

☐ c) R

☐ d) U

Solution:

For all the six companies, the sum of the percentage increases in the three years together is 30. In such cases, the company with the most evenly matched values for the three years will have the highest percentage increase i.e, company Q. Choice (B)

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If the sales of company R in 2015 were Rs.125 cr, what were the sales in 2018?

- ☐ a) Rs.158 cr
- ☐ b) Rs.162 cr
- ☐ c) Rs.165 cr
- ☐ d) Rs.168 cr

Solution:

The sales in 2018 = $125 \times 1.2 \times 1.1 \times 1.0 = 165$ cr

Choice (C)

In 2019, the sales of companies P and Q were Rs.260 cr each. What was the difference in their sales in 2017?

- ☐ a) 6.2 cr
- ☐ b) 8.6 cr
- ☐ c) 9.8 cr
- ☐ d) 11.5 cr

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Solution:

Let the sales of companies P and Q in 2017 be x and y respectively.

For company P, sales in 2019 = $x \times 1.15 \times 1.1 = 1.265x$

For company Q, sales in 2019 = $y \times 1.1 \times 1.2 = 1.32y$

Given that $1.265x = 1.32y = 260$ cr

$$x = \frac{260}{1.265} = 205.5 \text{ cr and } y = \frac{260}{1.32} = 197 \text{ cr.}$$

The difference = $205.5 - 197 = 8.5$ cr

Choice (B)

In 2019, the sales of company U were Rs.416 cr. What were its sales in 2016?

- ☐ a) Rs.268cr
- ☐ b) Rs.287cr
- ☐ c) Rs.291 cr
- ☐ d) Rs.300 cr

Solution:

Let the sales in 2016 be x.

The sales in 2019 would be $x \times 1.0 \times 1.01 \times 1.05 = 1.365x$

Given that $1.365x = 410$ cr

$$\therefore x = \frac{410}{1.365} = 300.4 \text{ cr.}$$

Choice (D)

By what percentage is the sales of company T in 2019 more than its sales in 2015?

- ☐ a) 72
- ☐ b) 66
- ☐ c) 60
- ☐ d) 55

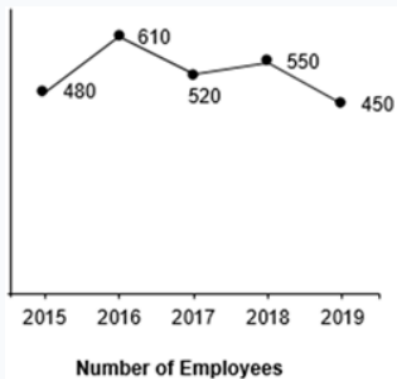
Solution:

Let the sales of company T in 2015 be x .
 The sales in 2019 would be $x \times 1.1 \times 1.15 \times 1.05 \times 1.25 = 1.66x$
 The required percentage = 66

Choice (B)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The line graph shows the number of employees and the sales per employee of a company across five years.



Profitability (%) =

$$\frac{\text{Sales} - \text{Expenditure}}{\text{Expenditure}}$$

$\times 100$

What was the percentage change in the sales of the company from 2016 to 2018?

- ☐ a) 2.3
- ☐ b) 2.9
- ☐ c) 3.5
- ☐ d) 4.2

Solution:

Sales of the company in 2016 = $610 \times 2.4 = 1464$

Sales of the company in 2018 = $550 \times 2.6 = 1430$

The percentage change = $\frac{34}{1464} \times 100 = 2.3$

Choice (A)

In which of the given years were the sales of the company the highest?

☐ a) 2016 (Correct Answer - ✓)

☒ b) 2017 (Your answer - ✗)

☐ c) 2018

☐ d) 2015

Solution:

The sales of the company was the highest in 2016 as the number of employees is the highest in that year and the sales per employee is very close to the highest value.

Choice (A)

If the profitability in 2017 was 10%, what were the expenses per employee in that year?

☐ a) 2.18 cr

☐ b) 2.25 cr

☐ c) 2.36 cr

☐ d) 2.45 cr

Solution:

The sales in 2017 = $520 \times 2.6 = 1352$.

The expenses in 2017 = $\frac{1352}{1.1} = 1229$

The value of expenses per employee = $\frac{1229}{520} = 2.36$ cr

Choice (C)

What was the average sales of the company for the given period?

- ☐ a) Rs.1252 cr
- ☐ b) Rs.1267 cr
- ☐ c) Rs.1281 cr
- ☐ d) Rs.1298 cr

Solution:

The average sales during the given period

$$\begin{aligned} &= \frac{480 \times 2.2 + 520 \times 2.6 + 520 \times 2.6 + 450 \times 2.3}{5} \\ &= \frac{1056 + 1464 + 1352 + 1430 + 1035}{5} \\ &= 1267.4 \end{aligned}$$

Choice (B)

What was the highest percentage change in the sales of the company from one year to the next?

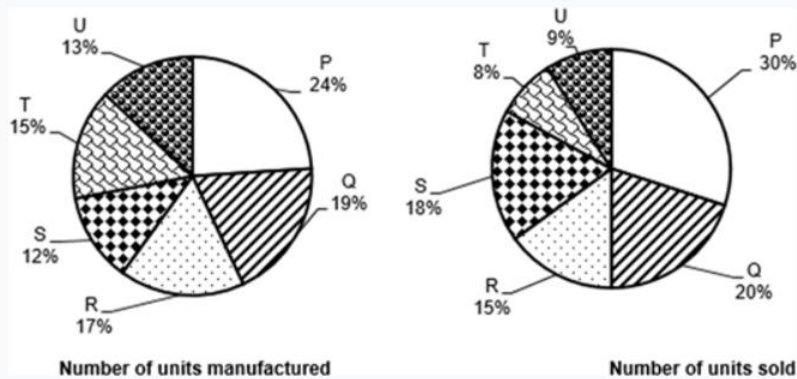
- ☐ a) 31.2
- ☐ b) 34.1
- ☐ c) 36.2
- ☐ d) 38.6

Solution:

The highest percentage change would be from 2015 to 2016 and it is $\frac{408}{1056} \times 100 = 38.6$
Choice (D)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The pie charts give the breakup of the number of units of all the six products – P, Q, R, S, T and U-manufactured and sold by a company ABC.



If the company manufactured not more than 500 units of all the six products together, at least what percentage of it was sold?

- ☐ a) 25
- ☐ b) 20
- ☐ c) 15
- ☐ d) None of these

Solution:

Maximum number of units manufactured = 500

Minimum number of units sold = 100

The required percentage = $\frac{100}{500} \times 100 = 20$

Choice (B)

If the company manufactured 3900 units of product U, at most how many units did it sell?

- ☐ a) 1200
- ☐ b) 1500
- ☐ c) 1800
- ☐ d) None of these

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

If the company sold 3200 units of product T, at least how many units of the product did it manufacture?

Solution:

As 3200 units of product T were sold, the total number of units sold = $\frac{3200}{8} \times 100$
= 40,000.

$\therefore$ At least 60,000 units were manufactured.

The number of units of product T manufactured = $\frac{15}{100} \times 60,000 = 9000$ Ans: (9000)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

If the company sold 4500 units of product R, at least how many units of product Q did it manufacture?

Solution:

If 4,500 units of product R were sold, the total number of units sold = 30,000.

The minimum number of units of product Q manufactured = $\frac{19}{100} \times 45,000$
= 8550.

Ans: (8550)